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# Artificial intelligence in SOC alert triage: the moderating role of cyber hygiene, Zero Trust and tool fragmentation

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Each line is an alert; only a few are signal.


## Abstract

The effect of artificial intelligence (AI) on alert triage in a Security Operations Center (SOC) depends on the quality of the environment in which it is deployed. This article proposes a conceptual framework for examining this thesis. Its research question is as follows: whether and how the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of AI, relative to conventional triage, on detection quality and on investigation cost.

The framework distinguishes information volume from information noise. It treats the use of AI as the focal variable, the three factors as moderators, and alert volume and rule configuration complexity as control variables. Four research propositions are formulated, each accompanied by its expected direction and its refutation criterion. A three-phase validation protocol is proposed: a reproducible targeted review, an exploratory qualitative study, and then a multilevel analysis complemented by a replay of labelled alerts.

The article reports no empirical results. It aims to provide a testable basis for determining under what conditions AI genuinely reduces information noise rather than automating existing deficiencies.


**Keywords:** SOC; alert triage; information noise; artificial intelligence; cyber hygiene; Zero Trust; tool fragmentation; moderation effect; human decision-making.

## 1. Introduction

SOCs concentrate the monitoring, detection, analysis and response functions for security incidents. They rely on solutions for security event management, endpoint detection, threat intelligence and response orchestration. The multiplication of cloud environments, the generalisation of digital identities and the growth of telemetry sources increase the volume of data to be processed, and organisations stack specialised tools to cover ever wider perimeters.

The literature describes the human consequences of this situation. Tariq et al. (2025) survey research on alert fatigue in SOCs. Alahmadi et al. (2022) and Kokulu et al. (2019) report, on the basis of interviews with analysts, the difficulties related to the relevance of alarms and to the fit between tools, processes and people. Vielberth et al. (2020) offer a systematic study of SOCs and their open challenges. Ofte and Katsikas (2023) examine operators' situational awareness.

Faced with these difficulties, AI is increasingly presented as a lever for prioritising, correlating and summarising alerts. Chhetri et al. (2024) study the cooperation between analysts and AI in this respect. However, AI is inserted into an environment that is already structured, with its detection rules, its tools and its security practices, and nothing allows us to assume that its contribution is identical from one SOC to another.

### 1.1 The gap addressed

The works cited most often examine alert load, human-AI cooperation or situational awareness. The relationship between the effect of AI and the quality of the technical and organisational environment is less formalised. The working hypothesis of this article is that this gap exists and that it is significant; the targeted review described in Section 5 must confirm or refute this before any submission.

A triage AI consumes context: asset identifiers, owners, account history, patch status, endpoint posture, logs from several tools. If this context is incomplete, inconsistent or scattered, the AI may go faster without deciding better. Conversely, a well-controlled environment may allow the AI to produce a real gain. Context is therefore not a mere confounding variable: it may condition the very effect one seeks to measure.

### 1.2 Research question

How do the maturity of cyber hygiene, the consistency of Zero Trust implementation and the degree of tool fragmentation modify the effect of artificial intelligence, relative to conventional triage, on detection quality and on investigation cost during alert triage in SOCs?

Three sub-questions break it down:

1. **SQ1 (hygiene).** Is the advantage of AI over conventional triage more pronounced when cyber hygiene is more mature?
2. **SQ2 (fragmentation).** Does this advantage diminish when tools are fragmented and poorly interoperable, or does AI on the contrary compensate for this fragmentation?
3. **SQ3 (Zero Trust).** For alerts related to identities and access, does a consistent Zero Trust implementation increase the advantage of AI?

Both outcomes are examined together: a time saving obtained at the price of missed threats does not constitute an improvement.

### 1.3 Contributions and outline

The article provides three elements: a functional definition of information noise distinct from volume and from alert fatigue; a moderation framework in which AI is the focal variable and three characteristics of the environment are moderators; and four falsifiable propositions with a validation protocol. Section 2 reviews the literature, Section 3 presents the framework, Section 4 formulates the propositions, Section 5 describes the validation method, and Sections 6 to 8 cover the discussion, the limitations and the conclusion.

## 2. Literature review

This review is deliberately focused on what underpins the model. It remains preliminary: Section 5 provides for a reproducible review before submission.

### 2.1 Alert volume and fatigue

Alert fatigue refers to the degradation of attention or vigilance that may appear when analysts handle numerous alerts whose relevance is low or difficult to establish (Tariq et al., 2025). The number of alerts measures load, not the difficulty of the work: two SOCs receiving the same number of alerts may face very different efforts depending on the available context, redundancy and investigation procedures. It is therefore necessary to distinguish the volume of events collected, the volume of alerts generated, the proportion of relevant alerts, the difficulty of qualification and the quality of final decisions.

### 2.2 AI in triage and human-AI cooperation

AI can prioritise alerts, link events, summarise context and automate repetitive tasks. Chhetri et al. (2024) argue for the value of a cooperation in which automation handles suitable tasks and human expertise remains engaged for ambiguous cases. Work on automation shows, however, that overreliance and complacency are known risks (Parasuraman and Riley, 1997; Parasuraman and Manzey, 2010). An AI that processes more alerts in less time therefore improves security only if the risk of missing a relevant threat does not increase. This caveat justifies measuring investigation cost and detection quality together. The available evaluations point the same way, with limits. A trial on eight tickets in a university SOC, run by the tool's own designers, reports faster triage in seven cases out of eight (Banstola et al., 2026). A multi-agent architecture reaches a higher F1 on actionable alerts than simple agents, with a median latency more than three times higher (Wei et al., 2025). A user study of six security orchestration, automation and response (SOAR) tools observes greater efficiency but lower accuracy and completeness of tickets (Bridges et al., 2023). A mixed-methods survey of 696 analysts shows that automation reduces monitoring and increases trust in its outputs, while systematic verification procedures mitigate this risk (Tilbury and Flowerday, 2026). Finally, a survey of 87 studies on alert filtering and triage finds that only nine rely on real SOC data or on a deployment, and that published results should be read as upper bounds (Ndichu et al., 2026).

### 2.3 Tool fragmentation and loss of context

SOCs combine several specialised technologies. This diversity improves coverage but may degrade integration: different identifiers for the same asset, redundant alerts, context to be reconstructed manually across several interfaces. Kokulu et al. (2019) document, on the basis of 18 interviews, mismatches between technologies, processes and people in SOCs; none of the SOCs concerned had a common interface bringing its tools together. The relevant variable is not the number of products, but interoperability, the semantic consistency of data and the sharing of context between components. An environment composed of many well-integrated tools is not fragmented in the sense of this model.

### 2.4 Cyber hygiene

Cyber hygiene encompasses the basic practices that keep an environment under control: asset inventory, patch management, administration of accounts and privileges, secure configuration. ANSSI (2017) offers a reference set of measures. For a SOC, these practices determine how easily an alert can be linked to an asset and to an owner, and the stability of baseline behaviours. An improvement in hygiene may nevertheless reveal previously invisible events and temporarily increase detections; the expected effect therefore concerns the interpretability of alerts, not their number. A conceptual analysis of SOC difficulties links the absence of maintained processes, incomplete documentation and insufficient information flows to rule tuning that can take three to six months (Onwubiko and Ouazzane, 2019); this is indirect evidence, not a measurement.

### 2.5 Zero Trust

Zero Trust rests on explicit verification of access, least privilege and the evaluation of context and risk at each request (Rose et al., 2020). The CISA maturity model (2023) structures its implementation into pillars, including identity and devices. The associated controls produce information on identities, devices and access conditions which, when correlated with other sources, can help qualify an access alert. But poorly harmonised policies may also multiply denials and context changes. Zero Trust is therefore neither an intrinsic source of noise nor a guarantee of noise reduction. Its role in this model is restricted to alerts related to identities and access, where it provides context.

### 2.6 Summary and current support

The following table indicates, for each element of the model, the assumed role and the degree of documentary support currently available to this article. It highlights the parts to be consolidated by the targeted review.

| Element | Assumed role in the model | Current documentary support |
|---|---|---|
| Alert fatigue and volume | Control variable | Strong (Tariq et al., 2025; Alahmadi et al., 2022) |
| Human-AI cooperation | Framework for assisted triage | Partial (Chhetri et al., 2024; Parasuraman and Manzey, 2010; Tilbury and Flowerday, 2026); evaluations mostly on public datasets (Ndichu et al., 2026) |
| Tool fragmentation | Moderator | Partial, mostly qualitative (Kokulu et al., 2019) |
| Cyber hygiene | Moderator | Reference frameworks (ANSSI, 2017), indirect evidence (Onwubiko and Ouazzane, 2019), little work on the link with the SOC |
| Zero Trust | Moderator, identity and access alerts | Reference frameworks (Rose et al., 2020; CISA, 2023), little work on the link with the SOC |
| Interaction between these factors and AI | Core of the contribution | No study found in a preliminary exploration; to be established by the targeted review |

The gap is therefore less the absence of literature on each of the factors than the absence, according to a preliminary exploration (October 2026) that did not query any academic database, of a framework that articulates them as moderators of the effect of AI on triage.

## 3. Conceptual framework

### 3.1 Definition of information noise

Information noise refers to the share of security information which, in a given operational context, mobilises analytical resources without contributing proportionately to the identification, qualification or management of relevant threats.

This definition is functional. A non-malicious alert is not necessarily noise: it may verify the coverage of a control or reveal a misconfiguration. Noise is judged in relation to the purpose of the investigation, the value of the signal and the effort required to exploit it. It is distinct from volume, which is a quantity, and from alert fatigue, which is a possible human consequence of repeated exposure to a load that is difficult to manage.

### 3.2 Model variables

The model poses a question of difference: does AI perform better or more cheaply than conventional triage, and does this difference vary with the environment? It retains one focal variable, three moderators, two outcomes and a set of control variables. The figure below gives the structure and the table the working definitions.

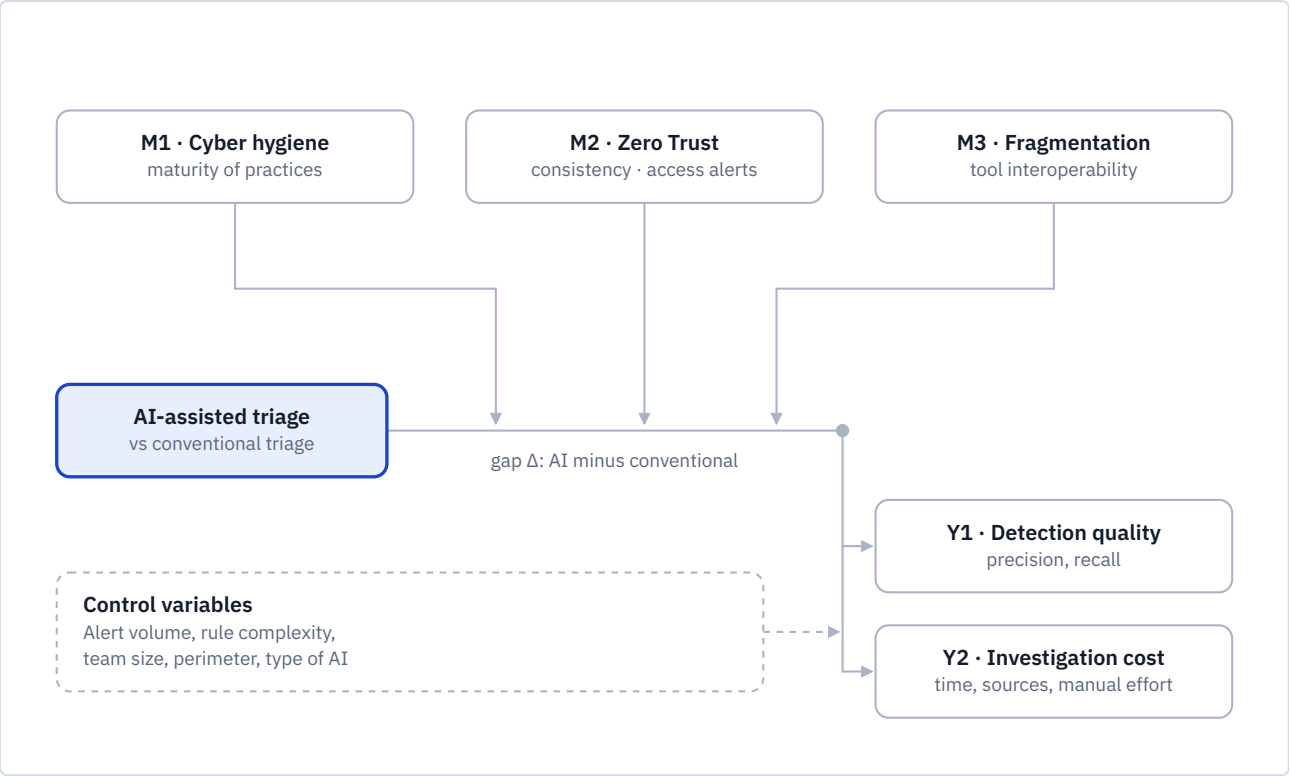



Figure 1. Moderation model: one focal variable, three moderators, two outcomes.

The three moderators act on the difference between AI and conventional triage, not directly on the outcomes; the control variables act on both conditions being compared.

| Role | Variable | Working definition |
|---|---|---|
| Focal | AI-assisted triage | Use of AI to prioritise, correlate or summarise alerts, compared with conventional triage based on rules and manual procedures. Described by its functions, its level of automation and human supervision. |
| Moderator M1 | Cyber hygiene maturity | Completeness of the asset inventory, patch compliance, control of accounts and privileges, maintained configurations. |
| Moderator M2 | Consistency of Zero Trust implementation | Harmonisation of access policies and integration of identity and endpoint data. Applied to alerts related to identities and access. |
| Moderator M3 | Tool fragmentation | Low effective interoperability: non-shared identifiers for the same asset, scattered context, number of interfaces to consult per alert. |
| Outcome Y1 | Detection quality | Precision, recall and false-negative rate on relevant alerts. |
| Outcome Y2 | Investigation cost | Qualification time, number of sources consulted, manual effort per alert. |
| Control | Volume and configuration | Alert volume, rule configuration complexity, team size and experience, monitored perimeter, type of AI. |

### 3.3 Reading the model

Let Δ be the performance difference between AI-assisted triage and conventional triage, measured separately on Y1 and on Y2. A moderator is a variable that makes Δ vary: the effect of AI is not assumed to be constant, it is assumed to be conditional. The model does not say that AI is effective or ineffective in general; it says where and under what conditions its effect changes.

Three clarifications frame this reading.

1. **Maturity and consistency are two distinct constructs.** Hygiene is measured by a maturity of practices, Zero Trust by the consistency of an architecture and its policies. They will be operationalised separately (Section 5).
2. **Volume and rule complexity are controls, not objects of study.** They influence alert load in both conditions being compared, but the framework does not seek to explain their own effect.
3. **Rival readings exist.** Hygiene may also act as a factor that modifies the number of alerts (direct effect), and the most mature SOCs may be those that adopt AI first (selection bias). The protocol in Section 5 provides for within-SOC comparisons to limit this second risk.

## 4. Research propositions

The following four propositions are theoretical statements to be tested, not results. Each specifies the expected effect, the mechanism and what would refute it. The "AI advantage" refers, for each outcome, to the difference favourable to AI relative to conventional triage: higher detection quality or lower investigation cost.

#### P1 Baseline effect: the gain concerns cost first

**Statement.** On average, AI-assisted triage reduces investigation cost (Y2) relative to conventional triage, but this reduction is not sufficient to establish an improvement in detection quality (Y1), whose difference may be zero, positive or negative.

**Mechanism.** AI acts first on the speed of qualification through prioritisation and summarisation. Detection quality depends on the accuracy of rankings and on the risk of masking an important alert, a risk known in decision-support systems (Parasuraman and Manzey, 2010). The available evaluations associate a gain in speed with lower accuracy and completeness of tickets (Bridges et al., 2023), which rules out inferring an improvement in Y1 from a decrease in Y2.

**Refutation.** P1 is refuted if cost is not lower under AI, or if the advantage on Y1 is uniformly positive across all environments, which would make it pointless to distinguish the two outcomes.

### P2 Cyber hygiene: the advantage grows with maturity

**Statement.** With alert volume controlled, the AI advantage on Y1 and on Y2 is greater the higher the maturity of cyber hygiene (M1).

**Mechanism.** A complete inventory, controlled accounts and stable configurations allow AI to link each alert to an asset and to an owner, and to rely on stable baseline behaviours.

**Caveat.** Better hygiene may reveal previously invisible events and temporarily increase the number of detections; hence the control for volume.

**Refutation.** The interaction between AI use and M1 is zero or of opposite sign after controlling for volume.

### P3 Tool fragmentation: the advantage decreases, or AI compensates

**Statement.** The AI advantage on Y1 and on Y2 is smaller the more tools are fragmented and poorly interoperable (M3).

**Mechanism.** Inconsistent identifiers and scattered context limit the ability of AI to link data; its outputs then rest on partial context.

**Rival proposition.** AI might on the contrary compensate for fragmentation by correlating scattered sources that the analyst would otherwise have to consult one by one. The advantage would then be more pronounced in a fragmented environment. These two readings are opposed and can be discriminated by the data; deciding between them is a possible contribution of the study.

**Refutation.** The interaction between AI use and M3 is zero or positive. A positive interaction would support the rival reading.

### P4 Zero Trust: the advantage grows with consistency, for access alerts

**Statement.** For alerts related to identities and access, the AI advantage on Y1 and on Y2 is greater the more consistent the Zero Trust implementation (M2), that is, the more policies are harmonised and identity and endpoint data are integrated.

**Mechanism.** Zero Trust controls produce usable context for qualifying an access alert: endpoint posture, session risk, access decision applied.

**Caveat.** Poorly harmonised policies may multiply denials and context changes, and reduce the contribution of Zero Trust to triage.

**Refutation.** No interaction for these alerts, or an interaction of the same magnitude for alerts unrelated to access. This second case would indicate that the construct measured reflects general maturity rather than Zero Trust.

## Summary

| Proposition | Factor | Expected effect on the AI advantage | Alerts concerned |
|---|---|---|---|
| P1 | None (baseline effect) | Advantage on Y2; advantage on Y1 not established | All |
| P2 | M1 Cyber hygiene | Grows with maturity | All |

| Proposition | Factor | Expected effect on the AI advantage | Alerts concerned |
|---|---|---|---|
| P3 | M3 Tool fragmentation | Decreases with fragmentation (rival reading: grows) | All |
| P4 | M2 Zero Trust | Grows with consistency | Identities and access |

The combination of the three moderators (higher-order interactions) is not formulated as a proposition: it is treated as an exploratory question in Section 7.

## 5. Proposed validation methodology

Validation follows a sequence of three phases, from the most exploratory to the most constraining. The same logic runs through it: comparing AI and conventional triage within contexts that differ in their hygiene, their Zero Trust and their fragmentation.

### 5.1 Phase 0. Reproducible targeted review

A restricted systematic review, reported according to PRISMA 2020 (Page et al., 2021), will verify the gap announced in Section 1.1. It will query the intersections between SOC, alert triage, AI, cyber hygiene, Zero Trust and tool integration, and will document the databases, the search strings, the inclusion criteria and the selection. Its result conditions the final formulation of the contribution.

### 5.2 Phase 1. Exploratory qualitative study

Semi-structured interviews will be conducted with SOC analysts, detection leads, CISOs and tool integration specialists. A target of around 12 to 20 interviews in 4 to 6 different SOCs (in-house, service providers, public and private sector) is proposed, until thematic saturation. The interviews will verify that the moderators correspond to real mechanisms and that the working definitions are operational, and will collect examples of cases where AI helped or hindered triage.

### 5.3 Phase 2. Multilevel analysis and controlled replay

**Multilevel analysis.** Alerts (level 1) are nested within SOCs or client perimeters (level 2). The moderators M1, M2 and M3 are measured at level 2; AI use varies within a single SOC where possible (before and after deployment, processing queues with and without AI). Mixed-effects models are estimated with cross-level interaction terms between level 1 and level 2 (Raudenbush and Bryk, 2002; Hayes, 2018). The power of level-2 interactions depends on the number of SOCs or perimeters, not on the number of alerts. If this sample size is insufficient to estimate three interactions, the study will retain two moderators, and managed service providers, which supervise several clients at different maturity levels, constitute a source of natural variation to be explored.

**Controlled replay.** A set of alerts labelled by experts serves as ground truth. Analysts process it with or without AI assistance, and the quality of the context provided is varied (complete or degraded inventory, shared identifiers or not, integrated identity data or not). This manipulation makes it possible to approximate a causal effect of the moderators on the AI advantage, which production data cannot establish on their own. Injected scenarios (for example derived from attack simulation exercises) complete the design to estimate recall, which is not observable in production. According to Ndichu et al. (2026), no public dataset combines real analyst verdicts, investigation metadata and longitudinal context; the replay set will therefore have to be built, or negotiated with partner SOCs.

**Exploratory calibration on GUIDE.** The public GUIDE dataset (Freitas et al., 2024) pairs SOC alerts with their analysts' verdicts (true positive, benign positive, false positive), but contains no measure of M1, M2 or M3, nor of investigation cost: it therefore cannot test P2 to P4. An exploratory examination of an extract of about 1,750 incidents from 480 organisations shows that verdicts cluster strongly by organisation (an intraclass correlation of about 0.7, estimated on organisations with at least two incidents, with wide uncertainty). This value supports the multilevel analysis and is a reminder that verdicts reflect both the environment and each SOC's grading practices, which these data alone cannot separate. It also suggests that the number of organisations, more than the number of alerts, will determine the power of the study.

### 5.4 Operationalisation of the indicators

Five rules apply to every indicator. They are fixed before data collection. The moderators (M1 to M3) are measured at the level of the SOC or perimeter, the outcomes (Y1, Y2) at the level of the alert. No moderator is computed from AI outputs or from alert volume, to avoid circularity. Each indicator draws on at least two independent sources (for example a declared and an observed inventory). Reliability is checked through inter-rater agreement (Cohen's kappa; Cohen, 1960) on audits and through sensitivity analysis to weighting.

**M1. Maturity of IT hygiene.** Hygiene is measured as a formative index (Diamantopoulos & Winklhofer, 2001): its components are not interchangeable, since a complete inventory does not compensate for uncontrolled accounts. Four components, each expressed as a proportion from 0 to 1 over the monitored perimeter:

| Component | Indicator | Source |
|---|---|---|
| Inventory | Share of observed assets (network, endpoint agents, directory) present in the official inventory with a named owner | Reconciliation of an observed scan with the inventory |
| Patching | Share of assets with no known exploited vulnerability (CISA KEV catalogue) beyond the delay set by the SOC's policy | Vulnerability management tool |
| Accounts and privileges | Share of privileged accounts reviewed in the last 90 days; share of dormant or service accounts without an owner | Directory, access reviews |
| Configurations | Share of assets compliant with a reference baseline (for example the CIS benchmarks) | Compliance tool |

The index is the unweighted mean of the four components, complemented by a "weakest link" version (minimum value) and by alternative weightings. A self-assessment score alone is set aside, because it tends to overstate maturity. Measurement is averaged over a quarter and taken before the alert observation window, since better hygiene can bring previously invisible alerts to light.

**M2. Consistency of Zero Trust.** The construct is the consistency of access policies and the integration of identity and device data, not the possession of products. Four indicators, drawn from the identity and device pillars of the CISA model (2023): share of accesses evaluated by a common policy decision point; share of applications covered by conditional strong authentication; share of endpoints whose posture is evaluated and transmitted; share of access events whose alert record automatically contains the endpoint posture and session risk. The number of distinct policy engines and of conflicting rules serves as an inverse indicator of consistency. M2 is checked as distinct from M1: their correlation is reported, and the models control for shared variance.

**M3. Fragmentation, not diversity, of tools.** Diversity is the number and variety of tools (D). Fragmentation is the effective lack of integration at the moment an alert is investigated. A SOC running twenty-five tools linked by common identifiers is diverse without being fragmented; a SOC with four unconnected tools is fragmented without being diverse. The indicators therefore concern the links, never a count:

| Indicator | Definition |
|---|---|
| Entity resolution | Share of the assets and identities in a sample of alerts found under a common identifier across all the sources consulted |
| Context accessibility | Number of distinct consoles to open to assemble a minimum context fixed in advance (asset, owner, user, history, vulnerabilities, neighbouring network activity), and the delay to obtain it |
| Automatic enrichment | Share of the minimum context elements already present in the alert record without manual action |

Two provisions protect the distinction. D is measured separately and entered as a control: the effect of M3 must persist at constant diversity, and a strong correlation between D and M3 would signal that the construct reduces to diversity. Second, M3 is measured on operation without AI, because an AI that enriches alerts itself masks the fragmentation. Underlying fragmentation, measured before AI, is distinguished from residual fragmentation, measured with AI; the gap between the two is the indicator of the compensation raised in P3.

**X. AI-assisted triage.** Status per alert or per processing queue (exposed to AI or not), complemented by a dose: assisted functions (prioritisation, correlation, summarisation, automated investigation), level of automation (recommends or decides) and share of alerts handled by AI. Assignment is controlled as far as possible: parallel queues, before and after deployment, or randomisation in replay.

**Y1. Detection quality.** Precision, recall and false-negative rate, reported by alert type and severity. Ground truth is established by double expert labelling, with arbitration of disagreements and a target kappa of at least 0.7. Analysts' production verdicts are not used as the sole ground truth: they are influenced by the AI itself and by each SOC's practices. Recall is estimated on injected scenarios.

**Y2. Investigation cost, net of verification.** The time saved thanks to AI may be taken back by the extra time spent verifying its outputs. A single clock, from opening to closing the alert, measures the balance but does not distinguish the two mechanisms and does not detect blind acceptance, which produces a short time without real gain. Cost is therefore decomposed: T = collection + analysis + verification + decision, then completed by a downstream cost.

| Indicator | Definition | What it separates |
|---|---|---|
| Net qualification time | Median time, log-normal distribution, by alert type | The overall balance |
| Collection and analysis time | Time before the AI recommendation is opened, or the conventional equivalent | The time saved |
| Verification time | Time spent on primary sources and evidence after exposure to the AI output | The time added |
| Correction rate | Share of AI recommendations modified or cancelled by the analyst | Calibrated trust |
| Rework rate | Share of alerts reopened or re-escalated within 7 days | The downstream cost of errors |
| Quality-adjusted net cost | Net time divided by the share of correctly qualified alerts | An apparent gain from a real one |

Durations come from interface log timestamps (opening of the recommendation, opening of sources, modification, decision), complemented by observed timing in replay. A perceived mental workload (short NASA-TLX; Hart & Staveland, 1988) may be added in replay. To detect over-reliance, known AI errors are injected into the replay set: if verification time falls and these false recommendations are accepted, the drop in Y2 reflects automation bias (Parasuraman & Manzey, 2010) rather than a gain. A gain is called apparent when net time falls alone, and net when the quality-adjusted cost falls without an increase in rework.

### 5.5 Rival hypotheses

Each proposition has at least one rival explanation that the data can arbitrate. The table pairs each with the design that allows it to be discriminated.

| Rival hypothesis | What it would predict | How to discriminate it |
|---|---|---|
| H1. Selection: mature SOCs adopt AI first | An apparent link between M1 and the advantage of AI that disappears once adoption is neutralised | Within-SOC comparisons, before and after, replay with random assignment |
| H2. Direct effect of hygiene on volume | M1 reduces the number of alerts in both conditions, without changing the gap between AI and conventional | Volume control; interaction distinguished from main effect; comparison at equal volume by strata |
| H3. Revelation: better hygiene first increases detections | A transiently higher volume in SOCs that are improving | M1 measured before the alert window, time series over several quarters |
| H4. Compensation: AI correlates scattered sources (rival of P3) | A positive AI × M3 interaction, and residual fragmentation below underlying fragmentation | Sign of the interaction; underlying versus residual gap |
| H5. Diversity rather than fragmentation | An effect attributed to M3 that vanishes when the number of tools D is controlled | Model with D and M3 together; comparison of coefficients |
| H6. General maturity rather than Zero Trust | An M2 interaction of the same size on alerts unrelated to access | Specificity test (access versus other alerts); M2 residual after M1 |
| H7. Apparent gain masking verification | Falling net time, with rising rework or acceptance of false recommendations | Decomposition of Y2, rework rate, injected errors |
| H8. Novelty and observation effect | A short-lived advantage at the start of deployment or in replay | Repeated measurement over several weeks, replay compared with production data |

### 5.6 Refutation criteria

Three verdicts are provided for each proposition: supported, refuted or inconclusive. The distinction is necessary because the number of SOCs limits level-2 power: a non-significant interaction refutes nothing. A proposition is refuted only if the 95% confidence interval of the effect lies entirely within an equivalence interval defined before collection (equivalence testing; Lakens, 2017), around a smallest difference of practical interest set in Phase 1 with practitioners. For example, a reduction in median qualification time of less than 15% might be judged to have no operational weight; this value is indicative and will be established in Phase 1. Without sufficient power, the verdict is “inconclusive”.

| Proposition | Supported if | Refuted if | Rival reading supported if |
|---|---|---|---|
| P1 | The quality-adjusted cost (Y2) is lower under AI, with no increase in rework, and the gap on Y1 differs from that on Y2 | The net effect on Y2 is equivalent to zero; or the gap on Y1 has the same sign and size as on Y2 in all contexts | The gain on Y2 disappears once verification and rework are included: the reduction was only apparent (H7) |

| Proposition | Supported if | Refuted if | Rival reading supported if |
|---|---|---|---|
| P2 | The AI × M1 interaction is positive after volume control, in within-SOC comparison, and robust to the index weightings | The interaction is equivalent to zero or negative after volume control | The interaction disappears when adoption is neutralised (H1) or volume is controlled (H2, H3) |
| P3 | The AI × M3 interaction is negative, at constant diversity D | The interaction is equivalent to zero | Positive interaction with reduced residual fragmentation (H4); or an effect that vanishes when D is controlled (H5) |
| P4 | The AI × M2 interaction is positive for access alerts and greater than for other alerts (triple interaction), after control of M1 | No interaction for access alerts; or an equivalent interaction for alerts unrelated to access | The interaction is found on all alerts and vanishes after control of M1 (H6) |

Propositions P2 to P4 are therefore never tested in isolation: each is confronted with its rival before being declared supported. The indicators, the equivalence thresholds and the analysis plan will be deposited before collection (preregistration). Results that contradict the propositions will be reported on the same footing as those that confirm them.

### 5.7 Analysis strategy

A descriptive analysis will characterise volumes, distributions and differences between contexts. The mixed-effects models will test the interaction between AI use and each moderator, with random effects by SOC and control variables; P4 will be estimated on identity and access alerts only, then compared with other alerts as a specificity test. Sensitivity analyses will vary the definition of the indicators, and multiple comparisons will be corrected. Since AI deployment is not randomised and the most mature SOCs may adopt it first, maturity is modelled explicitly and within-SOC comparisons are favoured. No result can be anticipated before data collection.

### 5.8 Ethics and confidentiality

Alert data are sensitive and may contain personal data. The study will require confidentiality agreements with the organisations, anonymisation or pseudonymisation of logs, and aggregated data for processing times so as not to evaluate analysts individually. Interviews will be subject to informed consent.

## 6. Discussion

### 6.1 Theoretical contributions

The framework shifts the research question from volume to the conditional value of AI. Instead of asking whether AI reduces alert fatigue, it asks in what context it improves detection and reduces investigation cost. This reformulation has three consequences.

1. **AI is treated as a processing capability embedded in an environment, not as an isolated solution.** Its effect is conditional, and it is this conditionality that is tested.
2. **A productivity gain is not a security gain.** By measuring Y1 and Y2 together, the framework prevents concluding that an AI performs well when it reduces processing time while letting more threats through.
3. **A rival reading is formulated from the outset.** For fragmentation, P3 and its rival proposition predict opposite signs, which makes the framework falsifiable on a point where practitioners'

intuitions diverge.

### 6.2 Implications for cyber governance

Security and SOC managers can draw from the framework a diagnostic grid before investing in an AI layer. Do the observed difficulties stem from poorly calibrated rules, an incomplete inventory, poorly controlled accounts, non-interoperable tools or a lack of procedures? If the answer lies in the environment, AI risks producing a speed gain without a detection gain, and making it harder to spot the deficiencies it automates.

The framework also suggests supplementing volume indicators with measures of relevance, context, qualification cost and decision performance, and reserving ambiguous cases for human supervision. It does not oppose innovation: it seeks to determine the conditions under which innovation produces a measurable improvement.

### 6.3 Sequencing of investments

If P2 and P4 are borne out, cyber hygiene and Zero Trust consistency are not merely stand-alone security objectives: they increase the return on investments in detection AI. If P3 is supported, tool integration precedes AI. If it is the rival proposition that prevails, AI may on the contrary be a short-term palliative for fragmentation. In both cases, the investment decision benefits from relying on a diagnosis of the context rather than on the tool's promise alone.

## 7. Limitations and future directions

### 7.1 Limitations

- **Status of the manuscript.** The article is conceptual. It rests on no original data and its propositions are not established. The literature review is preliminary and must be replaced by the reproducible review of Phase 0.
- **Construct validity.** Hygiene, Zero Trust consistency and fragmentation are composite constructs whose measurement depends on context. The proposed scales are inspired by reference frameworks (ANSSI, 2017; CISA, 2023) that were not designed to measure moderators of AI effectiveness and will need to be validated.
- **Scope of Zero Trust.** The moderator M2 applies only to identity and access alerts. It says nothing about the triage of other families of alerts.
- **Ground truth.** Precision and recall require a reliable reference on relevant events. In production, recall is not observable and relies on injected scenarios, whose representativeness is limited.
- **Selection and causality.** AI deployment is not randomised. Within-SOC comparisons and the controlled replay limit the bias without eliminating it.
- **Heterogeneity of AI.** Machine-learning classifiers, correlation engines and assistants based on language models have neither the same strengths nor the same weaknesses, and evolve quickly. The type of AI is a control, but its results might not transfer from one generation of tools to the next.
- **Generalisation.** In-house SOCs, service providers and public bodies differ in their resources and objectives. Comparisons must take this into account.

### 7.2 Future directions

- **Higher-order interactions.** Do the effects of the three moderators accumulate, offset one another, or is one of them sufficient? The question is exploratory and requires a larger sample than that of a

first study.

- **Temporal dimension.** Maturity evolves. A longitudinal study could track the AI advantage as hygiene, Zero Trust or tool integration improve.
- **Human factors.** The effect of human supervision, overreliance and complacency on Y1 could be modelled as an additional mediator or moderator. Systematic verification procedures, which mitigate the drop in monitoring (Tilbury and Flowerday, 2026), are a candidate moderator not retained here.
- **Adversary adaptation.** A triage AI may become a target. Extending the framework to robustness against attacks aimed at detection deserves dedicated work.

## 8. Conclusion

This article proposes to shift the analysis of alert fatigue from volume to the conditional value of artificial intelligence. It formulates a precise question: whether and how cyber hygiene, Zero Trust consistency and tool fragmentation modify the effect of AI on triage, in terms of detection quality and investigation cost.

The central proposition is that the operational value of AI in a SOC depends not only on its own capabilities, but also on the quality of configuration, integration and contextualisation of the environment in which it operates. Four falsifiable propositions, one of which pits two readings of the effect of fragmentation against each other, and a three-phase protocol make it possible to put it to the test.

If it is borne out, the question to ask before adding an AI layer is no longer only what the tool can do, but what the environment allows it to do. If it is refuted, the framework will at least have specified the conditions under which AI provides a measurable gain independently of context, which would also be a useful result.

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